Mechanistic Estimation of Neural Networks: Second Round Analysis

Q1: The Aggregate Response State (Gate-SVD Fitting)

Your intuition for the vector-metric family $T_{old} \approx \sum_{a=1}^{k} \text{sym3}(G_a \otimes v_a)$ is mathematically profound. It forms a perfectly closed state under the linear random-matrix transport $B = W\text{diag}(w_1)$, avoiding the Khatri-Rao rank explosion completely. The missing link is how to fit the incoming source at the age gate without materializing intractable tensors.

[Derived] The Exact Frobenius Fitting Procedure

At the age gate, a source tensor is $T = \sum_{j=1}^{n} w_{2,j}\text{sym3}(A_{:,j}A_{:,j}^T \otimes P_{:,j})$. Unfolding this symmetric tensor into an $n^2 \times n$ matrix yields $M = (A \odot A)\text{diag}(w_2)P^T$, where $\odot$ is the column-wise Kronecker product. Finding the optimal $\{G_a, v_a\}$ to minimize the Frobenius error is exactly equivalent to computing the truncated SVD of $M = \sum_a \text{vec}(G_a)v_a^T$.

Crucially, we can compute the top k components of M *without ever forming* M by using its $n \times n$ Gram matrix $S = M^T M$:

1. Form $G_A = A^T A$. (Cost: n^3 FLOPs $= 0.5$ units).
2. Form the elementwise square $K = G_A \circ G_A$. (Cost: negligible).
3. Form the $n \times n$ Gram matrix $S = P\text{diag}(w_2)K\text{diag}(w_2)P^T$. (Cost: n^3 FLOPs $= 0.5$ units).
4. Extract the top k eigenvectors $v_1, \ldots, v_k$ of S. (Cost: negligible).
5. The corresponding left singular matrices are algebraically $G_a = A\text{diag}(w_2 \circ (P^T v_a))A^T$. (Cost: $k \times n^3$ FLOPs $= k/2$ units).

Cost and k Expectation

- **Gate cost:** $\approx 1 + k/2$ units (paid once per source when it reaches age 4).
- **Transport cost:** $G_a \leftarrow BG_aB^T$ costs 2 units per component. $v_a \leftarrow Bv_a$ is negligible. Total old-tier cost: $2k$ **units per layer**.
- **[Conjecture] Expected k:** While $A \circ P$ has a flat spectrum, the Gram matrix S relies on $A^T A \circ A^T A$, whose rank is strictly bounded by the structural rank of P (~ 230 directions). Because S isolates the optimal Frobenius projection, its spectrum will decay sharply. I expect $k \in [3, 8]$ is sufficient for the 2% relative D_{21} error, putting the old-tier bill at 6–16 units per layer.

[Measure-this] Falsifier

Load a saved dense A, P, and w_2 at layer 4. Run the 5-step algorithm above for $k = 5$. Reconstruct the slice approximations $D_{21}^{approx} = \sum_{a=1}^{5}[(G_a \circ I)v_a \mathbf{1}^T + 2\text{diag}(v_a)G_a]$. Compare the relative RMS error against the exact $(A \circ P)\text{diag}(w_2)A^T$. This executes on a CPU in under 5 seconds.

Q2: Novel Formulations (Outside the Sum-of-Sources)

Given that expectation-only replacements, Tucker bases, and column-subsampling caps are closed, any valid formulation must transport the *realized odd parity state* linearly.

Proposal: Quenched Tensor-Sketching on the Birth Index

- **The Mathematical Object:** Instead of keeping sources separate or decomposing them via SVD, we track a *single* fixed-width synthetic source $\tilde{A} \in \mathbb{R}^{n \times d}, \tilde{P} \in \mathbb{R}^{n \times d}$.
- **[Derived] The Update Rule:** When a source hits the age gate, project its n birth dimensions down to d dimensions using a fixed, quenched Gaussian random matrix $\Omega \in \mathbb{R}^{n \times d}$:

  $$\tilde{A}_{new} = A\Omega, \quad \tilde{P}_{new} = P\text{diag}(w_2)\Omega$$

  Add this to the global synthetic source: $\tilde{A}_{old} \leftarrow [\tilde{A}_{old}, \tilde{A}_{new}]$ and $\tilde{P}_{old} \leftarrow [\tilde{P}_{old}, \tilde{P}_{new}]$. Transport is exactly $\tilde{A} \leftarrow B\tilde{A}$ and $\tilde{P} \leftarrow B\tilde{P}$.
- **Cost:** Transporting an $n \times d$ matrix is d/n units. If we sketch the *entire* aggregated old tier down to $d = 2048$ columns, the cost is **6 units per layer**.
- **Why it survives:** This uses a fixed Johnson-Lindenstrauss projection Ω on the birth index *before* Khatri-Rao expansion, preserving the bilinear geometry.
- **[Measure-this] Falsifier:** Load the sum of the exact D_{21}^{old} slices for all old sources at layer 14. Generate a single $\Omega \in \mathbb{R}^{n \times 2048}$. Compute sketched legs, sum their rank-2048 reconstructions, and check if relative error is $< 2\%$.

Q3: Hypothesis for the Leaders' Method

[Conjecture] Hypothesis: The leaders are executing exactly the **Gate-SVD Aggregate State (Vector-Metric Family)** derived in Q1, with $k \approx 4$.

The Argument

The leaderboard math leaves no other physically possible window. V29's young tier costs 143 units. The leaders score higher accuracy at 133–165 units, meaning the deep third cumulant costs 0 ± 15 units.

The Gate-SVD method strictly satisfies every constraint:

1. It is mathematically exact under random matrix transport.
2. It completely bypasses the $A \circ P$ Khatri-Rao flat spectrum by fitting the Frobenius norm of the source tensor *before* it is queried for D_{21}.
3. At $k = 4$, it costs exactly 8 units per layer, dropping their total bill to ≈ 151 units—dead center of the 133–165 unit window.

[Measure-this] The Observable Signature

If the leaders are using this, their computational cost will be *strictly flat* from layer 5 onward. Furthermore, if you inject a localized rank-1 perturbation into w_2 at layer 4 of a local run, the leaders' final mean vector error will be completely spanned by the top k left-singular vectors of the dropped residual $M - \sum_{a=1}^{k} \text{vec}(G_a) v_a^T$ at that specific age gate.